

Software Testing Lab Assignment

JUMBLE

Apply **mutation testing using Jumble** and evaluate the effectiveness of JUnit test cases.

For each problem:

1. Write the Java program as a separate class.
2. Write the corresponding JUnit test class.
3. Ensure that all original test cases pass.
4. Run Jumble on the Java class.
5. Record the:
 - Total number of mutants/mutation points
 - Number of killed mutants
 - Number of live mutants
 - Mutation score
6. Identify at least **one mutation that was not detected**, if applicable.
7. Add/improve test cases and run Jumble again. Compare the mutation scores.

Question 1 – Even or Odd

Write a Java method:

```
public static boolean isEven(int n)
```

that returns **true** if a number is even and **false** otherwise.

Write test cases for

- Positive even number
- Positive odd number
- Negative even number
- Negative odd number
- Zero

Run Jumble and record the mutation score.

Question 2 – Prime Number

Write a Java method:

```
public static boolean isPrime(int n)
```

that determines whether a given integer is prime.

Write test cases for

- 2
- 3
- 4
- 1
- 0
- A larger prime number
- A larger non-prime number

Run Jumble and record the mutation score.

Question 3 – Maximum of Three Numbers

Write a Java method:

```
public static int maximum(int a, int b, int c)
```

that returns the largest of three integers.

Test the following situations:

1. First number is largest
2. Second number is largest
3. Third number is largest
4. Two numbers are equal
5. All three numbers are equal
6. Negative numbers

Run Jumble and record the mutation score.

Question 4 – Grade Calculator

Write a Java method:

```
public static char calculateGrade(int marks)
```

using the following rules:

Marks	Grade
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90–100	A
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80–89	B
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70–79	C
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60–69	D
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Below 60	F
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Write test cases for:

- 95, 85, 75, 65, 50, Boundary values: 90, 80, 70, 60

Run Jumble and record the mutation score.

Question 5 – Discount Calculator

Write a Java method:

```
public static double calculateDiscount(double amount)
```

Example rules:

amount >= 5000 → 20%

amount >= 2000 → 10%

amount >= 1000 → 5%

otherwise → 0%

Create tests, run Jumble and record the mutation score.